

Solving Elliptical Diff's Equations

Suppose we want to solve the Poisson Equation

$$\nabla^2 V = - \frac{\rho(\vec{r})}{\epsilon_0}$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = - \frac{\rho(x, y)}{\epsilon_0}$$

Consider the problem of 4 conductive plates placed in the faces of a box, each plate with potential V_1, V_2, V_3, V_4 .

Hence, inside the box we do not have charge density and the Poisson Eq. becomes Laplace Eq.

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

To solve this equation, we make use of the numerical second derivative, hence,

$$\begin{matrix} U_{i,j} \\ \uparrow \quad \uparrow \\ x \quad y \end{matrix}$$

$$\frac{U_{i+1,j} + U_{i-1,j} - 2U_{i,j}}{(\Delta x)^2} + \frac{U_{i,j+1} + U_{i,j-1} - 2U_{i,j}}{(\Delta y)^2} = 0$$

We solve for $U_{i,j}$, Let's discretize the space in x and y equally, so $\Delta x = \Delta y$

$$\Rightarrow U_{i,j} = \frac{U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1}}{4} \quad (1)$$

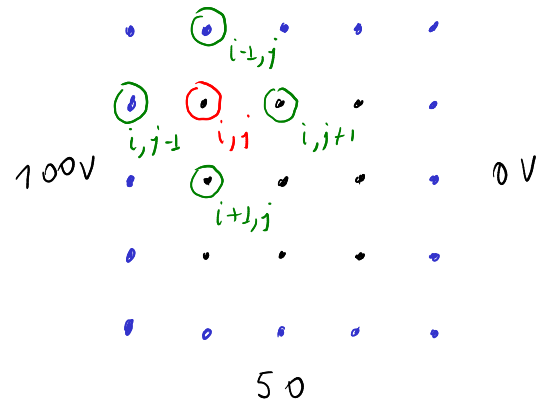
This is the fundamental relation to solve Laplace equation.

Let's assume that the area of interest is a square with area $A = 40 \text{ cm} \times 40 \text{ cm}$, and that,

$$U(0, y) = U_{1,j} = 75; U(40, y) = U_{N,j} = 50; U(x, 0) = U_{i,1} = 100; U(x, 40) = U_{i,N} = 0$$

To solve this equation Let's discretize the space by a grid of $N \times N$ points say $N=5$, as shown

We have the blue points as boundary conditions, and we aim to find the black points



Note that each point i,j corresponds to the average of the 4 nearest neighbors, as shown in Eq. 1,

hence the solution of the potential at the black points can be obtained by solving a linear system of equations given

$$\text{by, } \begin{bmatrix} -4 & 1 & 1 & 0 \\ 1 & -4 & 0 & 1 \\ 1 & 0 & -4 & 1 \\ 0 & 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} U_{22} \\ U_{23} \\ U_{32} \\ U_{33} \end{bmatrix} = \begin{bmatrix} -175 \\ -75 \\ -150 \\ -50 \end{bmatrix}$$

In this way we solve the differential equation,